Updated Technical Calculations for 1000W BLDC Motor

1. Vehicle Mechanical Calculations & Motor Selection

- **Gross Vehicle Weight (GVW):** 150 kg (including rider)
- **Rear Tire Radius (r):** 0.345 m
- **Tire Circumference (C):** C=2πr=2.168 mC = 2 \pi r = 2.168 \, \text{m}C=2πr=2.168m
- **Top Speed (V):** 60 km/h = 16.67 m/s
- **Acceleration:** 0–60 km/h in 8 s → a=2.083 m/s2a = 2.083 \, \text{m/s}^2a=2.083m/s2
- **Drag Coefficient:** Cd=0.9C_d = 0.9Cd=0.9, **Frontal Area:** A=0.8 m2A = 0.8 \, \text{m}^2A=0.8m2, **Air Density:** δ=1.2 kg/m3\delta = 1.2 \, \text{kg/m}^3δ=1.2kg/m3
- **Rolling Resistance Coefficient:** Cr=0.004C_r = 0.004Cr=0.004
- **Motor Efficiency:** ηm=0.85\eta_m = 0.85ηm=0.85

Forces on Flat Road:

- Rolling Resistance Fr=0.004×150×9.81=5.886 NF_r = 0.004 \times 150 \times 9.81 = 5.886 \, \text{N}Fr=0.004×150×9.81=5.886N
- Aerodynamic Drag Fd=12×0.9×1.2×0.8×(16.67)2=24.03 NF_d = \frac{1}{2} \times 0.9 \times 1.2 \times 0.8 \times (16.67)^2 = 24.03 \, \text{N}Fd=21 ×0.9×1.2×0.8×(16.67)2=24.03N
- Acceleration Force Fa=1509.81×2.083=31.85 NF_a = \frac{150}{9.81} \times 2.083 = 31.85 \, \text{N}Fa=9.81150×2.083=31.85N

Total Tractive Force (Flat):
Ft=5.886+24.03+31.85=61.77 NF_t = 5.886 + 24.03 + 31.85 = 61.77 \, \text{N}Ft =5.886+24.03+31.85=61.77N

Power Required (Flat Road):

P=Ft×Vηm=61.77×16.670.85≈1211.6 WP = \frac{F_t \times V}{\eta_m} = \frac{61.77 \times 16.67}{0.85} \approx 1211.6 \, \text{W}P=ηmFt×V=0.8561.77×16.67≈1211.6W

→ ✅ **Within the 1000 W Motor Range** if not climbing hills aggressively.

Hill Climb (30° slope, no acceleration):

Fhill=150×9.81×sin(30∘)=735.75 NF_{hill} = 150 \times 9.81 \times \sin(30^\circ) = 735.75 \, \text{N}Fhill=150×9.81×sin(30∘)=735.75N
Total Ft=735.75+24.03+5.886=765.67 NF_t = 735.75 + 24.03 + 5.886 = 765.67 \, \text{N}Ft

=735.75+24.03+5.886=765.67N
Required Power P=765.67×16.670.85=15.02 kWP = \frac{765.67 \times 16.67}{0.85} = 15.02 \, \text{kW}P=0.85765.67×16.67=15.02kW → ⚠️ **Too high for 1000 W motor**
→ Hill climb must be slow or avoided in steep inclines.

Torque:

- Wheel Torque T=Ft×r=61.77×0.345=21.3 NmT = F_t \times r = 61.77 \times 0.345 = 21.3 \, \text{Nm}T=Ft×r=61.77×0.345=21.3Nm
- Gear Ratio = 5 → Motor RPM = 2300, Wheel RPM = 460

🔋 Battery Pack Calculations (Based on 48 V, 40 Ah Lithium-ion)

Energy Requirement:

- Estimated energy = 50 km / 30 km/kWh = **1.667 kWh**

Battery Specs:

- Battery Voltage: 48 V
- Capacity Required: 1667/48=34.73≈40 Ah1667 / 48 = 34.73 \approx 40 \, \text{Ah}1667/48=34.73≈40Ah

Battery Cell Info:

- Cell Voltage = 3.6 V
- Cell Capacity = 2.5 Ah
- Energy/Cell = 9 Wh
- Cell Mass = 45 g

Battery Configuration:

- Series Cells (13S): 48/3.6≈1348 / 3.6 \approx 1348/3.6≈13
- Energy/String: 13 × 9 = 117 Wh
- Strings Needed: 1667/117≈151667 / 117 \approx 151667/117≈15
- Total Cells = 13 × 15 = 195
- Battery Mass = 195 × 45 g = **8.78 kg**

Peak Power:

- Peak Current/Cell = 5 A → 15 parallel → 75 A
- Peak Power = 75 × 48 = **3.6 kW**

✅ **Final Battery Pack:**

- **Voltage:** 48 V
- **Capacity:** 40 Ah
- **Energy:** 1.92 kWh
- **Mass:** ~9 kg
- **Configuration:** 13S15P

✅ **Final Motor and Battery Selection for Your Project**

Component	Specification
Motor	1000 W, 48 V BLDC, ~2300 RPM
Torque	21–25 Nm (flat), Peak ~100 Nm
Controller	1000 W compatible controller
Battery Pack	48 V, 40 Ah, 1.92 kWh, 13S15P Lithium-ion
Range	~50 km (flat road), moderate throttle

Would you like me to insert this section directly into your project report document or prepare a poster-friendly table as well?

You said:

Top Speed (V): 40 km/h

ChatGPT said:

Thanks! Based on your updated top speed of **40 km/h (11.11 m/s)**, here are the **revised calculations** for your EV conversion project using a **1000W, 48V BLDC motor**.

🔧 **Updated Technical Calculations – 1000W Motor at 40 km/h**

Vehicle Parameters

- **Gross Vehicle Weight (GVW):** 150 kg (bike + rider)
- **Wheel Radius (r):** 0.345 m
- **Top Speed (V):** 40 km/h = **11.11 m/s**
- **Drag Coefficient (Cd):** 0.9
- **Frontal Area (A):** 0.8 m²
- **Air Density (ρ):** 1.2 kg/m³
- **Rolling Resistance Coefficient (Crr):** 0.004
- **Motor Efficiency (η):** 85%

---

1. Forces Acting

a) Rolling Resistance:

Fr=Crr×m×g=0.004×150×9.81=5.89 NF_r = C_{rr} \times m \times g = 0.004 \times 150 \times 9.81 = 5.89 \, \text{N}Fr=Crr×m×g=0.004×150×9.81=5.89N

b) Aerodynamic Drag:

Fd=12·Cd·A·ρ·V2=0.5·0.9·0.8·1.2·(11.11)2=53.3 NF_d = \frac{1}{2} \cdot C_d \cdot A \cdot \rho \cdot V^2 \\ = 0.5 \cdot 0.9 \cdot 0.8 \cdot 1.2 \cdot (11.11)^2 \\ = 53.3 \, \text{N}Fd=21·Cd·A·ρ·V2=0.5·0.9·0.8·1.2·(11.11)2=53.3N

c) Acceleration Force (0–40 km/h in 6 sec):

a=11.116=1.85 m/s2Fa=m·a=150·1.85=277.5 Na = \frac{11.11}{6} = 1.85 \, \text{m/s}^2 \\ F_a = m \cdot a = 150 \cdot 1.85 = 277.5 \, \text{N}a=611.11=1.85m/s2Fa=m·a=150·1.85=277.5N

---

2. Total Tractive Force (Flat Road)

Ft=Fr+Fd+Fa=5.89+53.3+277.5=336.7 NF_t = F_r + F_d + F_a = 5.89 + 53.3 + 277.5 = 336.7 \, \text{N}Ft=Fr+Fd+Fa=5.89+53.3+277.5=336.7N

---

3. Power Requirement

P=Ft·Vη=336.7·11.110.85=4396.5 W≈4.4 kWP = \frac{F_t \cdot V}{\eta} = \frac{336.7 \cdot 11.11}{0.85} = \mathbf{4396.5 \, \text{W}} \approx 4.4 \, \text{kW}P=ηFt·V=0.85336.7·11.11=4396.5W≈4.4kW

⚠️ **Note:** This is peak power during acceleration. Your 1000W motor can handle this **briefly**. For cruising (no acceleration), the required power is:

Fcruise=Fr+Fd=5.89+53.3=59.2 NPcruise=59.2·11.110.85=774 WF_{\text{cruise}} = F_r + F_d = 5.89 + 53.3 = 59.2 \, \text{N} \\ P_{\text{cruise}} = \frac{59.2 \cdot 11.11}{0.85} = \mathbf{774 \, \text{W}}Fcruise=Fr+Fd=5.89+53.3=59.2NPcruise=0.8559.2·11.11=774W

✅ **1000W motor is sufficient for cruising at 40 km/h.**

---

4. Torque Calculations

At Wheel:

T=Ft·r=336.7·0.345=116.7 Nm(peak torque)T = F_t \cdot r = 336.7 \cdot 0.345 = \mathbf{116.7 \, \text{Nm}} \quad \text{(peak torque)}T=Ft·r=336.7·0.345=116.7Nm(peak torque)

Required Gear Ratio:
Assuming motor torque output = 20 Nm →
Gear ratio needed = 116.720≈5.8\frac{116.7}{20} \approx 5.820116.7≈5.8

So, a **6:1 gear ratio** is recommended.

✅ **Summary for 40 km/h Design:**

Parameter	Value
Motor	1000 W, 48 V BLDC
Max Speed	40 km/h
Cruise Power	~774 W (OK for 1000 W motor)
Peak Power	~4.4 kW (during acceleration)
Wheel Torque Req.	116.7 Nm
Gear Ratio	~6:1
Battery Needed	48 V, 40 Ah (1.92 kWh) for 50 km range